

An Atom–Photon Quantum Gate Using an Atomic Clock Qubit and a Birefringent Cavity

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Motivation

Quantum networks aim to link spatially separated quantum processors into a single, more powerful and more secure architecture by distributing entanglement between stationary qubits via flying qubits (in this case photons). Each network node must be able to perform three elementary tasks: store the incoming information of a photonic qubit onto a stationary qubits (storage), generate entanglement between the stationary and photonic qubit, and *process* quantum information locally through a deterministic quantum logic gate. The *CavityX* platform at the Max Planck Institute of Quantum Optics aims at realizing such a node using a single ^{87}Rb atom coupled to a birefringent fiber Fabry–Pérot cavity. It has previously demonstrated photon-to-atom state storage (94.7 % fidelity) and the generation of heralded atom–photon entanglement (87 % fidelity) – the first two node primitives. This thesis supplies the missing third primitive by realizing a deterministic atom–photon quantum gate, following the protocol proposed by Duan and Kimble [1].

A further design goal is robustness: qubits encoded in magnetically sensitive atomic states dephase due to ambient magnetic field fluctuations. This work instead encodes the atomic qubit in the magnetic-field-insensitive *clock states* of the ^{87}Rb $5S_{1/2}$ ground manifold, which is possible because – unlike in prior realizations of the Duan–Kimble protocol [2] – the required polarization selectivity is provided by the cavity’s birefringence rather than by Zeeman splitting.

Working Principle

The cavity’s two linear polarization eigenmodes are split by $\Delta_{\text{biref}} \approx 2\pi \times 505$ MHz. Only the π -polarized mode is resonant with the atomic π -transition. The orthogonal V -mode on the other hand is far detuned and reflects off the empty cavity regardless of the atomic state. A π -polarized photon reflecting off the atom-cavity system therefore picks up an atom-state-dependent phase: a π phase shift if the atom is in the uncoupled clock state $|0\rangle = |F=1, m_F=0\rangle$, and no phase shift if the atom is in the coupled clock state $|1\rangle = |F=2, m_F=0\rangle$ (Fig. 1). Because the atom is the control qubit and the photon polarization is the target qubit, this state-dependent reflection directly implements a controlled-phase-flip (CPF) gate in the photonic $\{|\pi\rangle, |V\rangle\}$ basis. Preparing the photon instead in a superposition of π and V – realized experimentally using circular polarization, $|R/L\rangle = (|V\rangle \pm |\pi\rangle)/\sqrt{2}$ – converts the same interaction into a controlled-NOT (CNOT) gate.

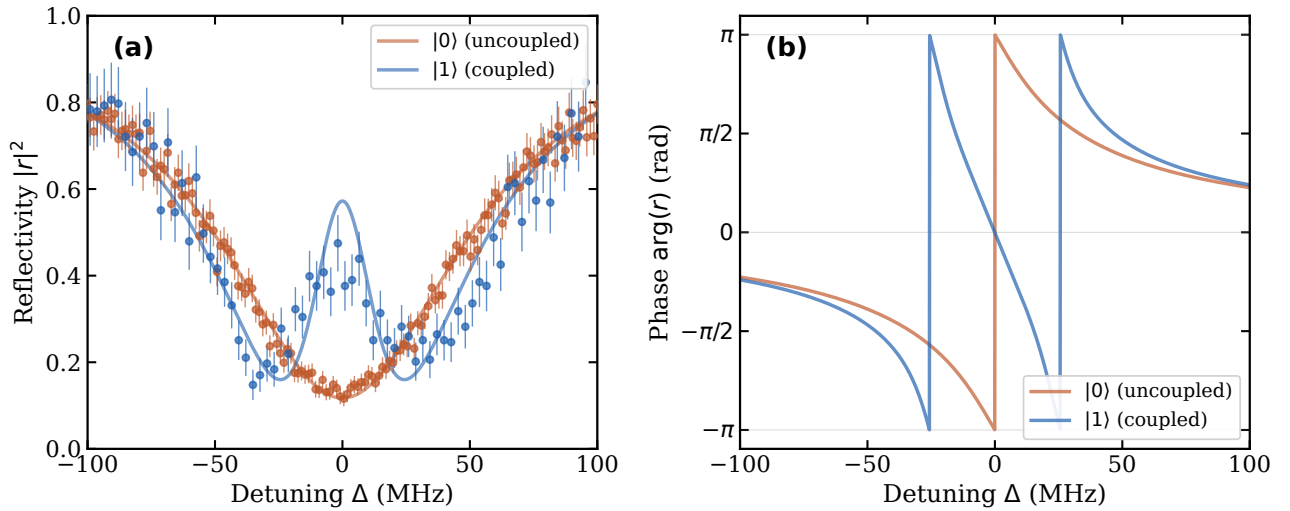


Figure 1: Normal-mode spectroscopy of the atom-cavity system. (a) Measured reflectivity $|r|^2$ versus probe detuning Δ for the atom in the uncoupled state $|0\rangle$ (red) and coupled state $|1\rangle$ (blue), with fits to the analytical model above; the coupled spectrum shows the vacuum Rabi splitting from which g is extracted. (b) Phase $\arg(r)$ computed from the same fitted model, showing the predicted π -phase-shift asymmetry between the two atomic states.

Because the V -mode reflects with near-unity amplitude while the π -mode reflectivity is reduced by intracavity losses, an array of Brewster windows was installed in the input path to attenuate the V -polarized component and balance the two reflection amplitudes, maximizing the CNOT gate overlap with the ideal truth table.

Experimental Results

The atom–cavity coupling strength was determined from normal-mode spectroscopy to be $g = 2\pi \times (25.9 \pm 0.5)$ MHz, consistent with a master-equation simulation used to identify the optimal operating point: a mean input photon number of $\bar{n} \approx 0.1$ and seven Brewster windows ($T_V \approx 24\%$). Operating at this point, the gate truth tables were measured by preparing the atom and photon in each of the four computational basis-state combinations and reading out the joint output state (Fig. 2). The measured truth tables overlap with the ideal CPF and CNOT gates by

$$\mathcal{O}_{\text{CPF}} = (92.41 \pm 0.52)\%, \quad \mathcal{O}_{\text{CNOT}} = (88.15 \pm 0.86)\%,$$

in good agreement with simulation. The dominant infidelity contributions are the residual asymmetry between the coupled and uncoupled π -mode reflectivities ($\sim 5\%$, set by the current cavity cooperativity) and imperfect preparation of the coupled atomic clock state ($\sim 6\%$).

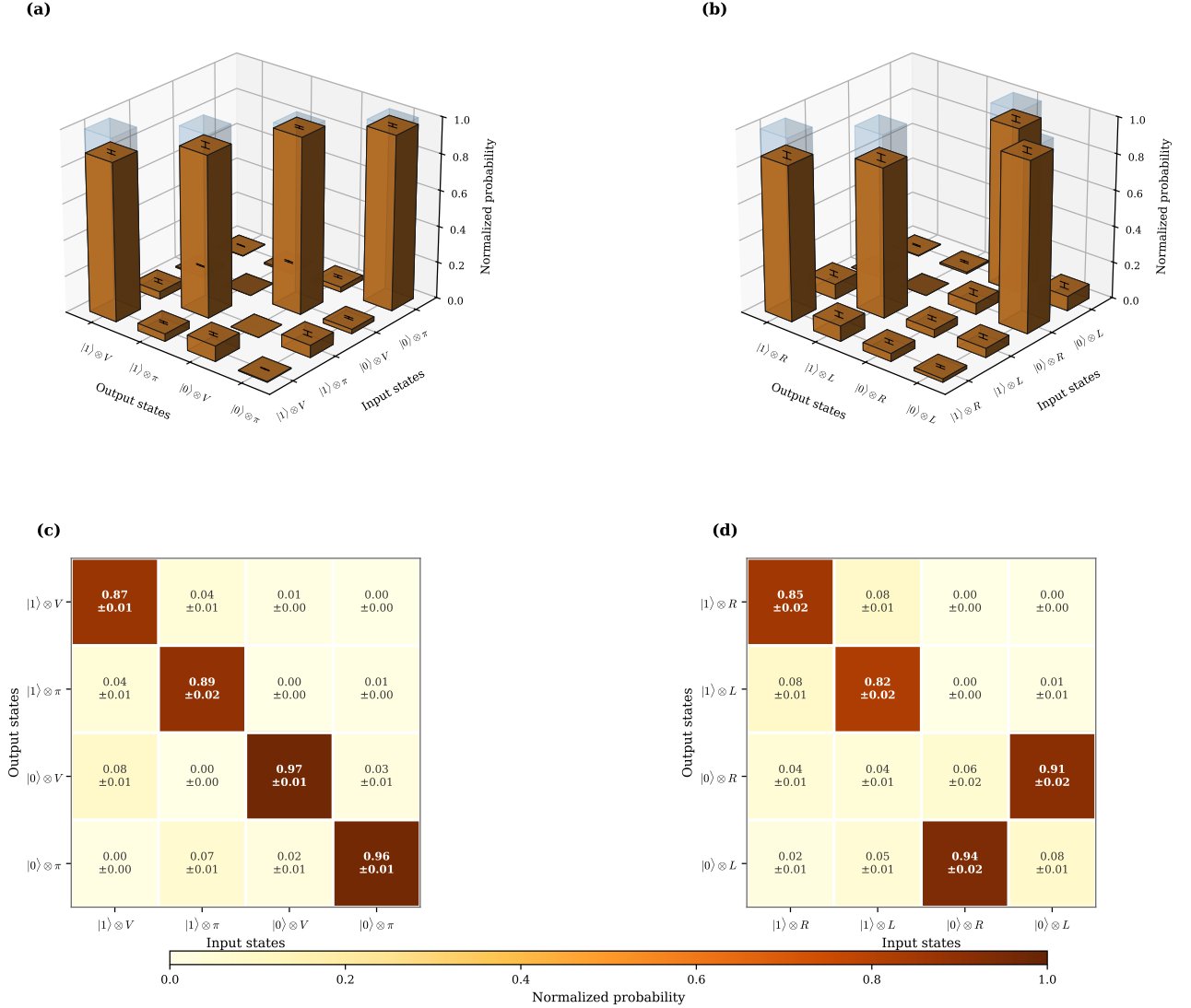


Figure 2: Measured gate truth tables at $\bar{n} \approx 0.1$: (c) CPF gate in the linear $\{|V\rangle, |\pi\rangle\}$ photon basis; (d) CNOT gate in the circular $\{|R\rangle, |L\rangle\}$ photon basis. Diagonal elements give the probability of the correct output state for each input combination.

To confirm the coherent, entangling action of the gate rather than merely its classical input–output behavior, an atom–photon Bell state $|\Phi^+\rangle$ was generated from a separable input and characterized via quantum state tomography, yielding a fidelity of $(65.90 \pm 1.73)\%$ – exceeding the 50% threshold required to certify entanglement, and thus confirming genuine atom–photon entanglement generated by the gate.

Conclusion and Outlook

This thesis experimentally realizes a deterministic atom–photon CPF/CNOT gate on the CavityX platform, completing the set of primitives – storage, entanglement generation, and now processing – required for a functional quantum-network node, while encoding the atomic qubit in magnetic-field-insensitive clock states for improved robustness. The next steps

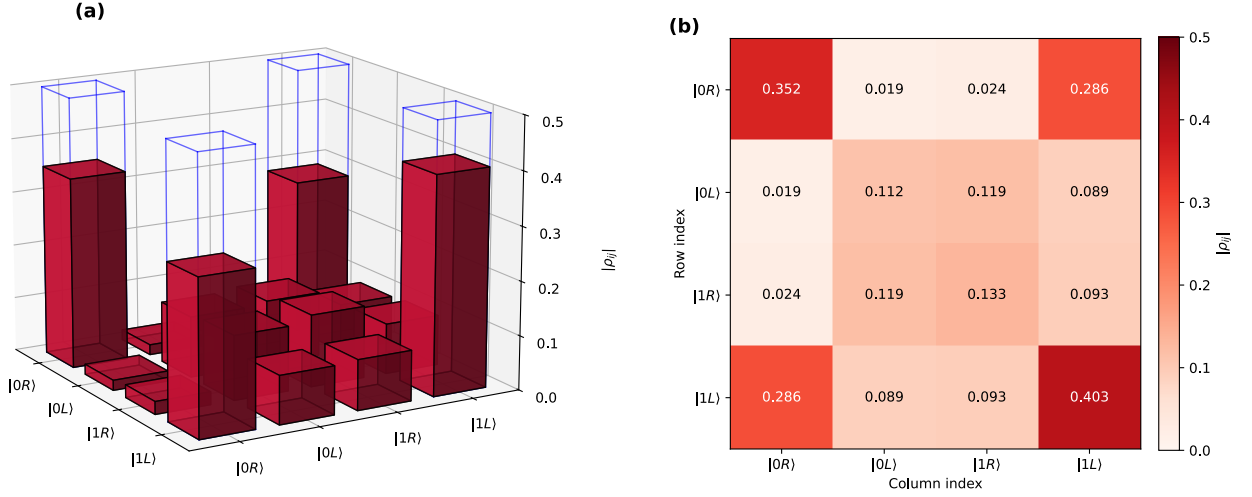


Figure 3: Magnitude of the reconstructed atom–photon density matrix $|\rho_{ij}|$ for the generated Bell state $|\Phi^+\rangle$, obtained via quantum state tomography. Population and coherence are concentrated on the $\{|0, R\rangle, |1, L\rangle\}$ subspace, as expected for $|\Phi^+\rangle$, with residual leakage into the wrong-parity states.

would be to systematically optimize the atomic state-preparation fidelity to improve the Bell-state fidelity, followed by inter-node protocols such as quantum state teleportation over the existing 24 km fiber link between MPQ Garching and LMU Munich [3]. Longer term, the same gate mechanism enables a photon–photon gate mediated by the single atom, and – by combining it with the platform’s quantum-memory capability – the heralded generation of entanglement between temporally separated photons.

References

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- [2] A. Reiserer, N. Kalb, G. Rempe, S. Ritter, *Nature* **508**, 237 (2014).
- [3] M. Büki *et al.*, *Phys. Rev. Lett.* **137**, 090803 (2026)